

PROJECT PROPOSAL

AI-Based Restoration of Damaged Brain Circuits and Digital Immortality

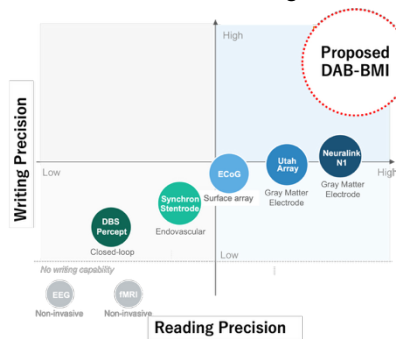
Unlocking the Primary/Ultimate Promise of Brain Machine Interfaces

EXECUTIVE SUMMARY

Every existing brain-machine interface can read neural signals with increasing resolution. None can write back with cell-level precision — and without precision writing, the primary clinical justification for invasive BMI remains unreachable: the partial replacement of destroyed brain circuits by AI. The DAB-BMI (Dissected Axonal Bundle Brain-Machine Interface) is the first neural interface architecture designed specifically to solve this problem, by targeting axons at the corpus callosum cross-section rather than stimulating gray matter. This program seeks funding to take the DAB-BMI from patented concept to a working closed-loop prototype in a non-human primate model, delivering a data package required for human trials for first-in-human studies in stroke and Alzheimer's disease — and opening, in the long term, a credible engineering path toward mind uploading.

1. THE PROBLEM EVERY BMI COMPANY HAS IGNORED

The global brain-machine interface industry — Neuralink, Synchron, Blackrock — is built on a single architectural assumption: that recording neural signals and routing them to external devices is sufficient to restore lost function. It is not. Every current BMI can **read** from the brain with increasing resolution. None can write back with precision.



The reason is fundamental, not incremental. Gray matter electrodes — the substrate of every existing BMI — face what Reid and colleagues have pointed out as the **"shotgun in the dark" problem**: electrical stimulation of gray matter inevitably activates axonal fibers of passage from distant, unrelated neurons interspersed among the local neuronal cell bodies being targeted. The result is an uncontrolled, diffuse activation pattern that bears no reliable relationship to the intended neural signal. Precise, cell-specific writing — the kind required to restore a lost sensory experience, reconstitute a damaged language circuit, or deliver targeted therapeutic neuromodulation — is architecturally impossible with gray matter electrodes.

This is not a problem that more electrodes, better algorithms, or higher-density arrays will solve. It is a **substrate problem**. The substrate must change.

The clinical consequence is direct. The single greatest argument for accepting the surgical risk of an invasive BMI — over non-invasive alternatives — is the possibility of **partial AI-based replacement of a damaged brain region**: restoring lost function not by routing signals around the lesion, but by substituting for the destroyed circuit directly. Without precision writing, this is impossible. A device that can read with high resolution but write only diffusely cannot replace a circuit — it can only observe one. Solving the precision writing problem is therefore not an incremental improvement to existing BMI technology. It is the prerequisite for unlocking the primary clinical justification for invasive BMI in the first place.

2. THE SOLUTION: PRECISION WRITING VIA THE DAB-BMI

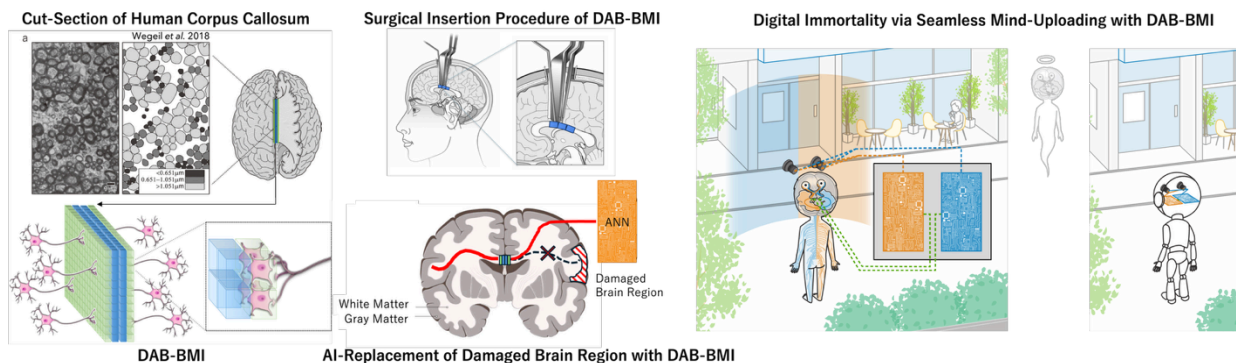
The Dissected Axonal Bundle Brain-Machine Interface (DAB-BMI) is the only neural interface in existence designed specifically to solve the precision writing problem. Rather than targeting gray matter — where neuronal cell bodies and axonal fibers of passage are irreducibly intermingled — the DAB-BMI targets **axons at the point of callosal severance**: the corpus callosum cross-section, where axonal fibers run in organized, geometrically defined bundles, separated from neuronal cell bodies entirely.

Why this eliminates the shotgun problem. At the corpus callosum cross-section, the stimulation target is exclusively axonal. There are no nearby neuronal cell bodies to activate spuriously, no fibers of passage from unrelated circuits, and no diffuse gray matter environment to contaminate the signal. Each stimulated axon belongs to a specific, identified long-range projection. This is, for the first time, a substrate that makes genuine cell-specific writing physically possible.

The DAB-BMI achieves this through three innovations that no existing device combines:

- **200-nanometer electrode spacing:** matched to the smallest axonal fiber spacing in the human and rodent corpus callosum. Every callosal fiber — every long-range interhemispheric projection — falls within the read/write coverage of the array. No gaps in signal coverage.
- **Genuine synaptic contact:** unlike proximity-coupled electrodes that influence neurons through electrical fields, the DAB-BMI establishes true biological-machine synaptic connections via biomolecular coating or cultured-neuron embedding directly on the CMOS surface. The signal is delivered, not broadcast.
- **Sub-2 ms bidirectional latency:** simultaneous action potential recording and electrical stimulation on every channel, with end-to-end latency within the biological spike timing window. The device communicates in the brain's own temporal language.

The DAB-BMI is built on a modified Neuropixels Ultra platform — a commercially validated, high-density probe with a proven chronic implant track record — surface-engineered for callosal insertion. The foundational hardware challenge is a materials science problem applied to an existing platform, not a ground-up device development problem.



3. CRITICAL GAP TO FILL IN

Big If True. The DAB-BMI's precision writing claim rests on one assumption not yet empirically demonstrated: that severed callosal axons will form stable, functional synaptic connections with the CMOS electrode surface. This is the program's central scientific bet — the most important unresolved question between the current concept and a working device. Naming it directly is not a weakness; it defines the experimental objective of Phase 1.

The Biological Challenge

When the corpus callosum is surgically severed, the exposed axonal stumps are not passive wires awaiting connection. They are living biological structures that respond actively to injury — retracting from the cut site, forming regrowing nerve fiber tips, triggering inflammation, and either initiating regenerative elongation or permanently breaking down, depending on the local molecular environment.

For the DAB-BMI to function, these severed axonal ends must contact the CMOS electrode surface and form stable, signal-transmitting synaptic connections — not merely electrical proximity, but the molecular machinery of a functional synapse, including the molecular machinery of a functional neuron connection. These connections must be maintained chronically under the mechanical and immunological conditions of a permanent implant. This has never been demonstrated at the scale and specificity the DAB-BMI requires.

Two candidate surface engineering approaches — biomolecular coating and cultured-neuron intermediary — are described in detail in the Appendix.

Why Now

CMOS electrode pitch is reaching axonal scale: Array density has advanced to where the 200-nanometer fiber spacing of the corpus callosum is now within reach of current fabrication processes.

Neurexin demonstrates silicon-axon synaptic contact: Immobilized neurexin has been shown to induce presynaptic differentiation on silicon substrates in vitro — the foundational biological assumption of the DAB-BMI is no longer speculative.

Gray matter BMIs have hit an architectural ceiling: The shotgun in the dark problem cannot be solved by more electrodes or better algorithms. Neuralink, Blackrock, and Synchron are all optimizing within this ceiling — creating a window for a substrate-level solution.

4. MARKET OPPORTUNITY

Primary Application: Neural Circuit Replacement

Every CNS disease treated by current medicine is managed symptomatically because no technology can replace the specific neural circuits destroyed by disease or injury. The DAB-BMI, combined with an AI circuit modeled on the brain's wiring, creates the first credible engineering pathway to physical circuit replacement. Clinical targets are large, underserved, and without disease-modifying alternatives:

- **Stroke-induced loss of cortical function:** 800,000 new strokes annually in the US alone; the majority of survivors live permanently with destroyed cortical circuits — language loss (aphasia), motor impairment, cognitive dysfunction — for which no restorative therapy exists. Neural circuit replacement directly addresses the pathology rather than compensating for it.
- **Alzheimer's disease:** the defining pathology is progressive circuit destruction. Hippocampal and entorhinal circuit replacement targets the mechanism of memory loss directly, not its downstream symptoms.
- **Traumatic brain injury:** focal circuit destruction with no current repair option. The TBI neurotechnology market is projected to exceed \$3 billion by 2030.

The combined addressable market for neural replacement therapy across these indications represents a multi-hundred-billion dollar opportunity — a market that does not yet exist because the enabling technology has not existed. The DAB-BMI creates it.

Secondary Applications : In Vitro Drug Screening, Peripheral BMI

In Vitro Drug Screening The DAB-BMI's fiber-level synaptic contact enables high-resolution measurement of drug effects on axonal conduction and synaptic transmission at single-fiber specificity — exceeding what conventional MEAs can achieve. The platform is directly applicable to CNS drug screening, tests for nerve cell damage, and validation of connection-promoting compounds including the biomolecular coatings developed within this program.

Peripheral BMI The DAB-BMI's axonal precision is directly transferable to peripheral nerve cross-sections, where bundled structure of nerve fibers creates a geometry analogous to the callosal target. This enables selective, fiber-specific stimulation and recording of individual sensory and motor axons, with applications in prosthetic limb control, pain modulation, and autonomic regulation. Every hardware advance made for the central application applies directly.

Ultimate Application: Mind Uploading and Digital Immortality

The same architecture that replaces a destroyed language circuit after stroke points, taken to its logical conclusion, toward something larger: the gradual, circuit-by-circuit replacement of the entire biological brain — preserving subjective experience throughout, because each artificial circuit is integrated before the corresponding biological one is lost. This is the technical definition of mind uploading, and the DAB-BMI is the only interface architecture capable of sustaining that continuity. Full discussion in the Appendix.

5. MILESTONES

The 36-month program advances the DAB-BMI to a validated closed-loop prototype in a rodent in vivo model.

Phase	Duration	Key metrics	End of phase / Kill switch
Phase 1 Surface engineering & in vitro	M1-18	Synaptic activity $\geq 80\%$ of sites; SNR $\geq 5:1$; viability $> 90\%$; $\geq 1,000$ ch bidirectional; write fidelity $\geq 80\%$; latency < 2 ms; stable 72h	Go: all metrics verified → proceed to animal studies Kill: transmission criteria unmet after 2 redesign cycles
Phase 2 Rodent in vivo safety	M19-23	Necrosis $< 5\%$; synaptic integrity $\geq 90\%$	Go: histology confirmed → proceed to closed-loop Kill: tissue or signal thresholds exceeded
Phase 3 Rodent closed-loop	M24-36	Closed-loop stable 240h; ANN fidelity $\geq 80\%$ of target neurons	Go: both criteria met → NHP pre-clinical package delivered Kill: instability or fidelity $< 80\%$ after optimization

6. ETHICAL FRAMEWORK AND REGULATORY STRATEGY

Animal Research — the Replace, Reduce, Refine ethical standard

- **Replace:** Phase 1 is conducted entirely in vitro. No animals are used until Phase 2.
- **Reduce:** Formal statistical power calculations determine minimum sample sizes. Rodent model precedes NHP to minimize higher-order animal use.
- **Refine:** All protocols developed under animal research ethics committee oversight. Humane endpoints, analgesia, and daily pain monitoring defined and approved before Phase 2 initiation.

Regulatory Pathway

- **Device classification:** DAB-BMI is a highest-risk FDA device category (hardware + biological component). preliminary meeting with the US device regulator (FDA) initiated at Month 6; parallel consultation with the Japanese device regulator (PMDA) at Month 12.
- **First-in-human entry point:** Patients with medically refractory epilepsy already scheduled for corpus callosotomy — a population for whom the surgical procedure is standard of care, minimizing incremental risk and providing the strongest benefit-risk framing for early human trials.
- **regulatory process for human trials:** The data package required for human trials delivered at program completion targets a formal preliminary FDA alignment meeting at program end, with human-trial device application for the first-in-human feasibility study to follow.

Toward Human Application: Social Readiness and Ethical Responsibility

The corpus callosotomy pathway is already standard-of-care surgery, making first-in-human trials a defined regulatory target. That proximity imposes two obligations. First, neural data at DAB-BMI resolution may carry cognitive content no prior technology could access. Governance frameworks — informed consent, encrypted storage, enforceable deletion rights — must be built in from the outset. Proactive engagement with emerging neurorights legislation is a program requirement. Second, the critical public distinction is architectural: the DAB-BMI reads and writes at identified axonal fibers in one anatomical location. It does not access the whole brain or eliminate autonomy. Communicating this clearly is both an ethical obligation and the precondition for social license.

APPENDIX

Two Candidate Surface Engineering Approaches

- **Approach A — Biomolecular coating:** The CMOS surface is functionalized with a combination of axon-guidance molecules (laminin, L1-CAM, netrin-1), connection-promoting factors (neuroligin-1, thrombospondin-1), and anti-inflammatory agents. The cocktail is designed to attract severed axonal regrowing nerve fiber tips toward the electrode, suppress glial scarring, and promote functional presynaptic terminal formation directly on silicon. This approach is scalable and substrate-agnostic. The key uncertainty is whether synaptogenesis on silicon can be induced reliably at 200-nanometer electrode spacing across the full array.
- **Approach B — Cultured neuron intermediary:** A monolayer of iPSC-derived neurons is cultured onto the CMOS array surface prior to implantation. These cells serve as biological intermediaries — already electrophysiologically active and capable of forming conventional biological synapses with incoming callosal axonal regrowing nerve fiber tips. The electrode communicates with the callosal axons through the cultured neuron layer rather than attempting direct axon-silicon contact. Biologically more conservative, but introduces questions of long-term cell viability, immune rejection, and interface stability.

Existing evidence and the precise gap. Synaptic contact between neurons and silicon has been demonstrated in model systems. Guided axonal regeneration toward functionalized surfaces has been demonstrated. The gap is achieving both simultaneously — at 200-nanometer electrode spacing, at the callosal cross-section, with chronic clinical stability. Both approaches run in parallel from Month 1; the Phase 1 Go/No-Go criterion measures directly whether either has succeeded before any animal procedures begin.

Mind Uploading and Digital Immortality — Extended Discussion

This is the technical definition of mind uploading. Not a sudden digital copy — which raises the philosophical problem of whether the copy is the same person — but a continuous, biological-to-artificial transition in which subjective experience is preserved throughout, because the artificial circuits are integrated into the living brain before the biological ones are lost. The DAB-BMI is the only interface architecture capable of sustaining that continuity: its precision writing capability is what ensures that the artificial circuit is experienced as *this person's own* sensation, not as a foreign signal grafted onto the brain.

The commercial and societal implications of genuine digital immortality — the preservation of personal identity, memory, and consciousness beyond biological death — are beyond the scope of any current market analysis. What can be said with precision is this: every step of the DAB-BMI's clinical development roadmap, from stroke rehabilitation to Alzheimer's circuit replacement, is also a step on the path toward mind uploading. The technology is not being developed for this purpose first; it arrives at this destination because the underlying engineering problem is the same throughout.

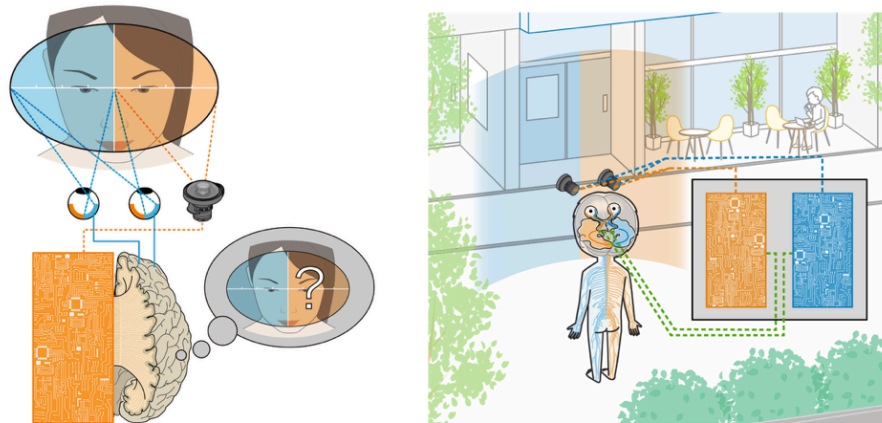


Figure 1: Subjective Test for Machine Consciousness

Artificial neural networks are connected with the biological hemispheres via the proposed DAB-BMI.

a) A conceptual depiction of the test. If “we experience the machine visual field, we must say that consciousness has emerged in the machine side given the “master-master” constraint of biological conscious vision (see [1]).

b) Actual depiction of the test. The cortical hemispheres are connected with spiking neural networks via DAB-BMI.

Phase 1 (Months 1–18) — Hardware Development & In Vitro Demonstration

Core objective: Demonstrate bidirectional, biologically faithful signal transmission across the surface-modified Neuropixels Ultra array in vitro, independently verified.

- **Task 1 — Surface engineering:** Neuropixels Ultra modification via two parallel approaches from Month 1. Iterative optimization of axon-guidance molecule cocktails, connection-promoting factors, and anti-inflammatory surface coatings. Parallel structure ensures a failure in one does not stall the program.
- **Task 2 — ANN construction:** Visual system AI circuit mimicking real neuron firing built on GPU cluster in parallel with hardware. Validated against published reference data from the brain's visual area before in vivo integration.
- **Task 3 — IV&V:** Independent verification by external laboratory (e.g., RIKEN CBS). Blinded data analysis and formal written report.

Phase 1 Go/No-Go (Month 18): In vitro bidirectional transmission $\geq 1,000$ channels, < 2 ms latency, write fidelity $\geq 80\%$, stable 72h — IV&V verified. AI circuit matches the brain's visual responses within 15%. No animal procedures until both criteria are met.

Phase 2 (Months 19–36) — Rodent In Vivo Integration & Closed-Loop Demonstration

Core objective: Demonstrate chronic DAB-BMI implantation safety in the rodent corpus callosum and closed-loop DAB-BMI $\leftrightarrow$ ANN operation in a live animal model.

- **Task 1 — Surgical safety (Months 19–28):** ethics-committee-approved rodent implant. Precision-guided surgical insertion into rodent corpus callosum; 30-day chronic maintenance with daily welfare monitoring, post-mortem tissue staining analysis.
- **Task 2 — Closed-loop integration (Months 25–36):** Full DAB-BMI $\leftrightarrow$ ANN system in rodent model. Real-time spike sorting, ANN-driven microstimulation, behavioral readout. Visual stimulation paradigm tests whether ANN-driven callosal stimulation produces measurable interhemispheric perceptual effects.
- **Task 3 — IV&V:** Tissue assessment without knowing the treatment group and spike data reanalysis. Rodent in vivo data package compiled for subsequent NHP program.

Final Exam (Month 36): (1) Safety: necrosis $< 5\%$, synaptic integrity $\geq 90\%$, no significant immune cell invasion — IV&V confirmed. (2) Closed-loop: stable 240h; ANN microstimulation induces intended firing patterns in $\geq 80\%$ of target neurons. Both met $\rightarrow$ program complete.

Competitive Position

- **vs. Neuralink / Blackrock:** both are gray matter electrode systems. Precision writing is architecturally impossible. The DAB-BMI's callosal substrate is categorically different.
- **vs. Synchron:** endovascular placement trades surgical risk for severely reduced channel density and zero precision writing capability. Complementary indication, not a competitor for circuit replacement.
- **vs. closed-loop DBS (Medtronic Percept):** onboard sensing enables crude state detection but stimulation remains diffuse gray matter broadcast. DAB-BMI enables true cell-specific closed-loop.

Why this PI. Masataka Watanabe, Ph.D. is the sole inventor of the DAB-BMI (patent filed). His research spans mathematical neuroscience (*Neural Networks*, 2001), human psychophysics (*Journal of Vision*, 2008), non-invasive brain imaging (*Science*, 2011), and invasive in vivo electrophysiology (*Nature Communications*, 2024) — the exact combination required to execute this program. No other investigator holds both the hardware IP and the pre-clinical methodological toolkit.

Investment and Future Vision

The Ask

We are seeking funding to execute the 36-month program described above, delivering a working closed-loop NHP prototype, an regulatory data package required for human trials, and a defensible IP portfolio covering the DAB-BMI hardware, biocompatible surface chemistry, and closed-loop ANN architecture. This positions the program for a Series A raise targeting first-in-human feasibility studies in refractory epilepsy patients already scheduled for corpus callosotomy — the optimal first clinical population — with a direct line to pivotal trials in stroke and Alzheimer's disease.

Cost structure: NHP animal research is the largest single driver — non-negotiable for IND-enabling pre-clinical data and not replaceable by rodent models. Hardware development requires parallel iterations of both surface engineering approaches before the Phase 1 Go/No-Go criterion is met. Both are time-dependent; the 36-month program timeline is the minimum credible window, not a conservative estimate.

IP position: the DAB-BMI corpus callosum insertion architecture is patented. The biocompatible surface chemistry is independently licensable to the broader electrode manufacturing market. The closed-loop ANN-BMI architecture generates further filings at Phase 1 exit.

Beyond Neural Repair: A Long-Term Vision

Precision writing at the axonal level has one implication that extends beyond medicine: it is the only substrate capable of supporting the gradual, circuit-by-circuit replacement of the biological brain — the technical definition of mind uploading. The PI is the originator of the Subjective Test of Machine Consciousness and author of *From Biological to Artificial Consciousness* (Springer). The same investigator who holds the DAB-BMI hardware IP holds the verification framework for machine consciousness. The convergence is not coincidental.

Contact: Masataka Watanabe, Ph.D. | Principal Investigator | BiTS Accelerator Demo Day

Key References: [1] *Neural Networks* 14(4–5), 2001 [2] *J. Vision* 8(11), 2008 [3] *Science* 334(6057), 2011 [4] *Nature Comms.* 15(1), 2024 [5] Springer book: *From Biological to Artificial Consciousness*